

TDV 2215 Terminal - Complete Analysis

Self-Test Memory Addresses - Renamed & Documented

Boot Progress & Status Tracking

Address	New Name	Usage	Description
0x5FFD	boot_progress_marker	Boot sequence tracking	Values: 0x234 (ROM check), 0x24A (hardware test), 0x24D (init complete)
0x5FFB	current_function_marker	Debug tracking	Tracks current operation for debugging (0x25C, 0x261, 0x26D, etc.)
0x5E0C	system_status_word	Cold/warm boot detection	0x8080 = warm boot (skip tests), other = cold boot (full test)

Memory Test Variables

Address	New Name	Usage	Description
0x6000	display_memory_start	Video RAM base	Start of 8KB display buffer (0x6000-0x7FFF)
-	Address-in-address patterns	RAM integrity	Low/high byte address written to each location for verification

NVRAM Test Variables

Address	New Name	Usage	Description
0x5E03	nvram_test_address_pointer	NVRAM scanning	Current test address (0x50-0x5F config range)
0x5E01	nvram_checksum_accumulator	Data integrity	Accumulates checksum for NVRAM verification

Z80SIO Serial Port Test Variables

Address	New Name	Usage	Description
0x5D52	z80sio_channel_a_test_status	Host port test	Channel A (host communication) test results
0x5D54	z80sio_channel_b_test_status	Keyboard port test	Channel B (keyboard interface) test results
0x5D51	serial_transmit_retry_counter	TX reliability	Retry counter for serial transmission failures
0x5D2E	tx_buffer_counter	Buffer management	Transmit buffer byte counter

Display & Timing Control

Address	New Name	Usage	Description
0x5DF9	display_refresh_control	Video timing	Controls display memory updates and VSYNC
0x5E21	interrupt_activity_flag	IRQ coordination	Set by interrupt handlers, cleared by main loop
0x5E25	timer_channel_flags	Multi-timer control	Manages multiple timer channels (bits 0-7)
0x5E22	display_timeout_counter	Screen timeout	30000-count display timeout timer

String Data - Renamed & Documented

System Identification Strings

Address	New Name	Content	Usage
0x221E	terminal_model_string	"TDV 2215"	System identification display
0x2229	firmware_revision_string	"Rev. Lev. 11 /"	Firmware version information

Error & Status Messages

Address	New Name	Content	Usage
0x0286	error_unknown_string	"???"	Unknown error placeholder
0x028A	error_number_prefix_string	"ERROR NO.:"	Error message prefix
0x294D	nvramp_error_message	"Error in EAROM, call System Operator."	NVRAM failure message

User Interface Strings

Address	New Name	Content	Usage
0x1E0E	push_key_menu_title	"PUSH-KEY MENU"	Configuration menu header
0x1E1C	hex_code_prompt_string	"HEX CODE : "	Hex input prompt
0x1C50	cursor_position_debug_string	"Cursor position :"	Debug/status display
0x39BC	configuration_menu_title	"CONFIGURATION"	Main config menu title

Control Characters & ESC Sequences

Address	New Name	Content	Usage
0x35F0	ESC_character	0x1B	ESC character for escape sequences
0x35F1	CSI_bracket_character	0x5B ('[')	CSI sequence bracket character

Major Functions Analysis - Renamed & Documented

Character Processing & Terminal Emulation

Original Name	New Name	Functionality
<code>FUN_ram_2e8c</code>	<code>convert_ascii_hex_to_binary</code>	Converts ASCII hex chars ('0'-'9','A'-'F','a'-'f') to binary values
<code>FUN_ram_22fb</code>	<code>escape_sequence_parser</code>	Main ESC sequence parser entry point
<code>FUN_ram_2316</code>	<code>control_character_handler</code>	Handles control chars (0x11=XON, 0x13=XOFF)
<code>FUN_ram_25a6</code>	<code>control_code_dispatch_table</code>	Jump table dispatcher for control codes (0x00-0x1F)
<code>FUN_ram_241b</code>	<code>execute_control_character</code>	Executes control character functions
<code>FUN_ram_2557</code>	<code>terminal_character_output</code>	Main character output handler with flow control

Cursor & Display Management

Original Name	New Name	Functionality
<code>FUN_ram_31e9</code>	<code>cursor_movement_handler</code>	Handles cursor positioning and screen navigation
<code>FUN_ram_3675</code>	<code>write_character_with_cursor_wrap</code>	Character output with automatic line wrapping
<code>FUN_ram_368c</code>	<code>check_auto_wrap_condition</code>	Determines when to wrap cursor to next line
<code>FUN_ram_33db</code>	<code>get_default_tab_stop_count</code>	Returns tab stop spacing (default 1 or custom)

Terminal Mode & Configuration

Original Name	New Name	Functionality
<code>FUN_ram_3fc6</code>	<code>configure_terminal_mode_settings</code>	Complex mode configuration using lookup tables
<code>FUN_ram_3755</code>	<code>initialize_terminal_display_mode</code>	Initialize display parameters and attributes
<code>FUN_ram_37a4</code>	<code>set_display_attribute_bit</code>	Set individual display attribute bits
<code>FUN_ram_2b35</code>	<code>calculate_character_attribute_code</code>	Calculate character rendering attributes

Configuration Data Structures

Address	New Name	Purpose
0x0AA0	<code>terminal_config_data_table</code>	Main configuration parameter table
0x387E	Configuration lookup table	Mode setting validation table

Terminal Emulation Capabilities Detected

Control Character Support (0x00-0x1F)

- XON/XOFF Flow Control:** 0x11 (XON), 0x13 (XOFF) detected in control handler
- Jump Table Dispatch:** 32-entry table for all control characters

- **Bell Function:** String "Bell" found, likely 0x07 (BEL) support

ESC Sequence Processing

- **ESC Character:** 0x1B stored at 0x35F0
- **CSI Support:** '[' character (0x5B) stored at 0x35F1 for ESC[sequences
- **Parser Architecture:** Dedicated escape sequence parser function
- **State Machine:** Multi-stage parsing with character classification

Display Features

- **Cursor Control:** Advanced cursor positioning with wrap detection
- **Character Attributes:** Attribute calculation and bit manipulation
- **Auto-wrap:** Configurable line wrapping behavior
- **Tab Stops:** Configurable tab stop handling

Terminal Modes & Configuration

- **Mode Tables:** Complex lookup table system for terminal modes
- **NVRAM Storage:** Persistent configuration in EAROM
- **Flow Control:** Hardware and software flow control support
- **Multi-mode Support:** Different terminal emulation modes

Character Set Support

- **ASCII Processing:** Full ASCII character processing
 - **Hex Input:** ASCII hex to binary conversion for setup modes
 - **Graphics Characters:** References to "Graphic" mode in strings
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Hardware Interface Architecture

Serial Communication (Z80SIO)

- **Channel A:** Host computer communication (primary data)
- **Channel B:** Keyboard interface
- **Channel C:** Printer interface
- **Flow Control:** DTR/RTS hardware handshaking
- **Test Patterns:** 0x55, 0x2A loopback verification

Display System

- **Video Memory:** 8KB buffer at 0x6000-0x7FFF

- **VSYNC Timing:** Interrupt-driven display refresh
- **Character Attributes:** Bit-mapped attribute system
- **Cursor Management:** Hardware cursor with blink control

Configuration Storage

- **NVRAM/EAROM:** 16 bytes (0x50-0x5F) for settings
 - **Checksum Validation:** Data integrity verification
 - **Default Fallback:** Built-in defaults if NVRAM fails
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Error Handling & Diagnostics

Self-Test Error Codes

- **0x12:** DTR/RTS signal failure
- **0x13:** Serial pattern transmission failure
- **0x14:** Serial loopback failure
- **0x19:** NVRAM checksum failure
- **0x1A:** NVRAM not responding
- **0xB0+addr:** Specific NVRAM location failure

Debug Features

- **Progress Markers:** Boot sequence tracking
 - **Function Markers:** Current operation identification
 - **Status Words:** System state validation
 - **Error Messages:** User-friendly error reporting
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Key Insights

1. **Professional Terminal:** Full VT100-class terminal emulation capabilities
2. **Robust Design:** Comprehensive self-test and error recovery
3. **Configurable:** Extensive NVRAM-based configuration system
4. **Multi-Interface:** Host, keyboard, printer interfaces with flow control
5. **Standards Compliant:** ESC/CSI sequence support for terminal compatibility
6. **Industrial Grade:** Error reporting and diagnostic capabilities

The TDV 2215 represents a sophisticated terminal controller suitable for professional computer systems, with comprehensive terminal emulation, robust hardware management, and extensive configuration capabilities typical of high-end terminals from the 1980s era.

